

SUMTER, SC, USA

WMO: 720611

Lat: 34.000N

Lon: 80.367W

Elev: 181

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00202

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.4	25.5	7.0	7.9	29.9	11.8	10.1	35.5	19.0	60.6	17.3	56.4	2.4	350	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	99.0	81.5	95.2	78.3	92.9	77.4	89.6	93.5	83.6	89.1	79.5	89.7	6.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
88.3	206.6	93.1	81.6	165.7	87.4	77.1	142.2	83.1	55.3	93.6	47.7	94.7	43.2	89.6	95.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	14.0	12.1	15.2	100.8	4.9	4.1	11.7	103.8	8.8	106.2	6.0	108.5	2.4	111.4		
			14.4	84.0	4.7	4.8	11.0	87.4	8.3	90.2	5.6	92.9	2.2	96.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	45.4	48.6	55.1	63.9	72.3	79.5	81.5	80.8	76.8	64.7	53.3	47.6
	DBStd	15.29	10.69	10.05	9.47	7.59	6.00	4.76	3.49	4.52	6.56	7.79	8.07	8.66
	HDD50	628	223	137	57	3	0	0	0	0	0	3	51	154
	HDD65	2548	613	469	329	112	17	0	0	0	2	101	363	542
	CDD50	5808	79	97	214	419	690	884	978	955	805	459	149	79
	CDD65	2255	4	9	21	78	242	434	513	490	357	93	12	2
	CDH74	22716	27	70	248	795	2246	4552	5314	5005	3550	836	65	8
	CDH80	10426	3	8	37	206	893	2221	2607	2495	1729	223	4	0
Wind	WSAvg	4.3	4.5	5.2	5.6	5.6	4.6	4.0	3.9	3.2	3.6	3.7	3.7	4.2
Precipitation	PrecAvg	47.2	4.2	3.7	4.3	3.1	3.5	5.5	5.4	5.0	3.8	2.9	2.7	3.1
	PrecMax	67.7	8.3	7.6	11.7	9.1	8.1	14.3	13.8	11.5	9.7	15.4	8.7	7.3
	PrecMin	35.7	0.9	1.0	0.5	0.1	0.5	1.5	1.1	0.8	0.0	0.0	0.5	0.5
	PrecStd	8.1	2.0	1.8	2.3	1.9	1.7	2.8	2.6	2.4	2.2	2.8	1.7	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.5	81.1	83.8	88.3	95.2	102.3	99.4	103.9	105.7	89.7	80.6	73.5
		MCWB	67.0	65.8	64.7	70.4	75.0	81.0	77.7	89.1	90.8	74.8	69.2	63.3
	2%	DB	72.2	75.2	80.9	85.2	90.9	96.8	96.7	97.2	96.7	84.5	75.2	70.4
		MCWB	62.9	64.3	64.3	67.5	72.1	78.3	77.2	81.0	84.4	72.0	66.3	63.1
	5%	DB	67.7	71.8	76.7	82.0	88.3	93.1	93.5	93.3	91.4	81.9	72.5	66.3
		MCWB	61.0	61.2	62.3	67.2	71.9	76.6	77.6	78.7	78.8	69.7	65.1	60.7
	10%	DB	63.6	66.0	72.3	79.1	85.6	90.7	91.2	90.9	89.9	78.9	69.6	63.4
		MCWB	58.0	58.2	60.1	65.1	71.3	76.1	77.5	78.5	77.9	68.6	63.1	58.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	69.5	68.6	74.5	78.4	83.7	81.4	92.7	92.7	77.9	73.4	68.0
		MCDB	73.6	75.6	77.0	82.7	87.1	97.1	90.9	99.3	99.5	83.3	74.1	71.8
	2%	WB	65.3	66.6	66.9	72.2	75.8	80.8	80.3	89.5	90.4	75.9	71.0	64.4
		MCDB	70.3	74.1	74.8	79.4	85.0	91.9	90.6	93.2	95.0	81.6	72.5	66.7
	5%	WB	62.6	63.5	65.3	70.3	74.3	78.8	79.2	82.3	86.0	73.9	66.6	62.4
		MCDB	66.3	67.4	73.6	77.0	83.1	89.5	89.8	88.8	89.4	78.8	71.5	65.6
	10%	WB	58.9	59.7	63.0	68.4	72.8	77.6	78.4	79.0	78.2	71.7	63.2	59.3
		MCDB	62.6	64.3	68.9	74.9	81.3	87.6	88.1	88.2	86.2	76.9	67.0	62.6
Mean Daily Temperature Range	5% DB	MDBR	21.6	22.7	23.8	24.5	21.8	20.9	20.0	19.6	20.7	23.2	24.0	20.9
		MCDBR	26.3	27.0	28.4	26.6	26.0	23.4	22.2	22.0	23.9	25.2	25.7	25.1
		MCWBR	16.6	16.4	14.1	12.2	9.9	8.0	7.5	7.7	9.5	12.5	15.9	18.4
	5% WB	MCDBR	20.0	24.2	23.1	20.3	20.6	21.5	19.7	19.5	17.3	18.8	19.8	22.3
		MCWBR	15.4	16.0	12.8	11.2	9.0	8.0	7.7	8.4	8.6	10.5	13.9	17.9
Clear-Sky Solar Irradiance	taub	0.319	0.328	0.363	0.410	0.474	0.505	0.552	0.521	0.438	0.393	0.343	0.315	
	taud	2.500	2.472	2.400	2.259	2.130	2.086	1.977	2.054	2.274	2.369	2.469	2.549	
	Ebn at Noon	283	292	289	279	261	252	239	244	261	265	271	279	
	Edh at Noon	27	31	36	43	49	52	57	52	40	34	28	24	
All-Sky Solar Radiation	RadAvg	880	1081	1427	1796	1952	2030	1935	1752	1506	1261	967	773	
	RadStd	47	113	109	127	153	141	121	87	129	151	76	69	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (63 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+335	+238	

Nomenclature: See separate page